

Problem (informal): On the pure two-qubit state space $\mathbb{CP}^3$ with the Fubini–Study metric, study the spectrum and time-evolution of a weighted Hodge–de Rham Laplacian (on appropriate forms) with a thermal weight derived from a two-qubit Hamiltonian. Interpret the low-lying modes and dynamics as describing how probability distributions over pure states thermalize and how distinguishability/entanglement structure behaves under these dynamics.

SECTION 1 – THE GEOMETRIC SETUP OF THE PROBLEM.

We start by defining our Manifold. Let $\mathcal{M} = \mathbb{CP}^3$, the set obtained by taking $\mathbb{C}^4 - \{0\}$ modulo the equivalence relation:

$$z \sim w \Leftrightarrow z = \lambda w, \lambda \in \mathbb{C}^*$$

Which is the set of complex lines in $\mathbb{C}^4$ through the origin, of course under the usual projection topology that allows us to define the following chart

The standard atlas. For each $k = 0, 1, 2, 3$ define the open set

$$U_k = \{[z_0 : z_1 : z_2 : z_3] \in \mathbb{CP}^3 : z_k \neq 0\}.$$

So on each U_k , we're sure that at least the $z_k \neq 0$, which allows us to define the map.

$$\begin{cases} \phi_k : \mathcal{M} \rightarrow \mathbb{C}^3 \\ p = [z_i] \rightarrow \phi_k(p) = (w_i)_{i \in \{0,1,2,3\} - \{k\}} \end{cases} \quad \text{with } w_i = \frac{z_i}{z_k}$$

It is important to note that $\mathcal{M}$ isn't an ordinary manifold, it's a *Kähler* manifold:

A **Kähler manifold** is a complex manifold whose metric is simultaneously Riemannian, Hermitian, and symplectic, all in a compatible way; the **Kähler form** is the 2-form that encodes this metric plus the complex structure, and its being closed is what makes the manifold “Kähler.”

A Riemannian metric g on M is **Hermitian** if it is compatible with J , meaning

$$g(JX, JY) = g(X, Y) \text{ for all tangent vectors } X, Y.$$

The pair (g, J) gives an almost Hermitian manifold. From this data you define the associated 2-form

$$\omega(X, Y) = g(JX, Y), X, Y \in TM.$$

- ω is automatically nondegenerate and of type $(1, 1)$ (in complex terms), so it is a candidate symplectic form. [wikipedia+1](#)

The manifold is **Kähler** if, in addition, this 2-form is closed:

$$d\omega = 0.$$

So a Kähler manifold is equivalently:

- A complex manifold with a Hermitian metric whose associated 2-form ω is closed.
- Or a symplectic manifold (M, ω) equipped with an integrable complex structure J such that $\omega(\cdot, J\cdot)$ is a Riemannian metric

This triple (g, J, ω) is the Kähler structure, and $\mathbb{CP}^n$ with the Fubini–Study metric is the standard metric, which we’ll be taking for this problem.

The round metric on $\mathbb{S}^{2n+1}$ descends to a homogeneous and isotropic Riemannian metric on $\mathbb{CP}^{n+1}$, called the Fubini-Study metric.

On a complex manifold with local holomorphic coordinates $z_1, \dots, z_n$, the Kähler metric g is encoded by a function $\rho(z, \bar{z})$ such that

$$\omega = \frac{i}{2} \partial \bar{\partial} \rho$$

is the Kähler 2-form, and its components $g_{i\bar{j}}$ are

$$g_{i\bar{j}} = \frac{\partial^2 \rho}{\partial z_i \partial \bar{z}_j}.$$

This function ρ is called a **Kähler potential** for ω (and for the metric g).

On the affine chart $Z_0 \neq 0$ with $w_i = Z_i/Z_0$ for $i = 1, 2, 3$:

- The Kähler potential is

$$\rho(w, \bar{w}) = \log(1 + |w_1|^2 + |w_2|^2 + |w_3|^2).$$

- The metric components are

$$g_{i\bar{j}}(w) = \frac{\partial^2 \rho}{\partial w_i \partial \bar{w}_j}.$$

- The Kähler form is

$$\omega = \frac{i}{2} \sum_{i,j} g_{i\bar{j}} dw_i \wedge d\bar{w}_j.$$

Chart U_0 : $Z_0 \neq 0$

- Coordinates: $w_1 = Z_1/Z_0, w_2 = Z_2/Z_0, w_3 = Z_3/Z_0$.
- Potential: $\rho_0 = \log(1 + |w_1|^2 + |w_2|^2 + |w_3|^2)$.
- Metric: $g_{ij}^{(0)}(w) = \frac{\delta_{ij}}{1 + |w_1|^2 + |w_2|^2 + |w_3|^2} - \frac{\bar{w}_i w_j}{(1 + |w_1|^2 + |w_2|^2 + |w_3|^2)^2}, i, j = 1, 2, 3$.

Chart U_1 : $Z_1 \neq 0$

- Coordinates: $u_0 = Z_0/Z_1, u_2 = Z_2/Z_1, u_3 = Z_3/Z_1$.
- Potential: $\rho_1 = \log(1 + |u_0|^2 + |u_2|^2 + |u_3|^2)$.
- Metric: $g_{pq}^{(1)}(u) = \frac{\delta_{pq}}{1 + |u_0|^2 + |u_2|^2 + |u_3|^2} - \frac{\bar{u}_p u_q}{(1 + |u_0|^2 + |u_2|^2 + |u_3|^2)^2}, p, q \in \{0, 2, 3\}$

And so on and so forth...

now that we set up our Metric, we'd want to work with its volume form.

Recall The Kähler form is

$$\omega_{\text{FS}} = \frac{i}{2} \sum_{i,j=1}^3 g_{ij}(w) dw_i \wedge d\bar{w}_j.$$

On any **oriented Riemannian manifold** (M, g) , the volume form is the unique top-degree form dvol_g such that for any oriented orthonormal basis $(e_1, \dots, e_n)$ of $T_p M$ you have

$$\text{dvol}_g(e_1, \dots, e_n) = 1.$$

In local coordinates, this condition forces

$$\text{dvol}_g = \sqrt{\det(g_{ij})} dx^1 \wedge \dots \wedge dx^n.$$

So “volume form = $\sqrt{\det g}$ times coordinate wedge” is not a choice; it's the unique form that makes the integral of 1 equal the geometric volume.

On a **Kähler manifold** of complex dimension n :

- You have a Hermitian metric g_{ij} and its associated Kähler form

$$\omega = \frac{i}{2} \sum_{i,j} g_{i\bar{j}} dz_i \wedge d\bar{z}_j.$$

- In an orthonormal unitary frame, ω looks like

$$\omega = \sum_{k=1}^n e^k \wedge f^k$$

where $\{e^k, f^k\}$ is an orthonormal real coframe (think $\Re dz_k, \Im dz_k$).

Then the wedge power satisfies

$$\omega^{\wedge n} = n! \, e^1 \wedge f^1 \wedge \cdots \wedge e^n \wedge f^n,$$

which is exactly $n!$ times the Riemannian volume form.

So the *definition* becomes

$$\mathrm{dvol}_g = \frac{1}{n!} \omega^{\wedge n}$$

In these coordinates, that becomes

$$\mathrm{dvol}_{\mathrm{FS}} = \frac{1}{3!} \left(\frac{i}{2}\right)^3 \det(g_{i\bar{j}}(w)) \, dw_1 \wedge d\bar{w}_1 \wedge dw_2 \wedge d\bar{w}_2 \wedge dw_3 \wedge d\bar{w}_3.$$

***Tangent vectors as infinitesimal state changes:**

Take a normalized state $|z\rangle \in \mathbb{C}^4$ (Hilbert space), and its ray $[z] \in \mathcal{M}$.

By definition, $[z]$ is the equivalence class of all nonzero vectors proportional to $|z\rangle$, i.e. $\lambda |z\rangle$ with $\lambda \in \mathbb{C}^\times$; physically that's “same state up to global phase and normalization.”

To see what a tangent vector at $[z]$ means:

- Think of the quotient map $\pi: \mathbb{C}^4 \setminus \{0\} \rightarrow \mathcal{M}$, $\pi(|\psi\rangle) = [\psi]$.
- A tangent vector $X \in T_{[z]}\mathcal{M}$ can be represented by a derivative of a curve of states

$$t \mapsto |\psi(t)\rangle, |\psi(0)\rangle = |z\rangle,$$

modulo changes along the fiber (i.e. multiplying by a phase).

Concretely, one standard description is:

$$T_{[z]}\mathcal{M} \cong \frac{\{v \in \mathbb{C}^4: \langle z | v \rangle = 0\}}{\text{phase directions}}.$$

Intuitively:

- A vector v orthogonal to $|z\rangle$ describes an infinitesimal motion in Hilbert space that changes the ray nontrivially.
- Adding a term proportional to $|z\rangle$ (e.g. $i\alpha |z\rangle$) only changes the global phase or normalization, i.e. it moves along the fiber of π and does **not** change the point $[z]$ in $\mathcal{M}$.

If a small Hamiltonian H generates Schrödinger evolution,

$$|\psi(t)\rangle = e^{-itH} |z\rangle,$$

then at $t = 0$ the derivative is

$$\frac{d}{dt} |_{t=0} \psi(t) = -iH |z\rangle.$$

We decompose $-iH |z\rangle$ into a part parallel to $|z\rangle$ (global phase and normalization) and a part orthogonal to $|z\rangle$. The orthogonal part is precisely the tangent vector in $T_{[z]} \mathbb{CP}^3$, i.e. the actual **direction of state change modulo global phase**

1-form A as a quantum current / cost functional

A 1-form $A \in \Omega^1(\mathbb{CP}^3)$ assigns to each point $[z]$ a linear functional

$$A_{[z]}: T_{[z]} \mathbb{CP}^3 \rightarrow \mathbb{C}.$$

So for a tangent vector $X \in T_{[z]} \mathbb{CP}^3$, the value $A(X)$ is:

- A scalar depending linearly on the direction X .
- Hence a **directional cost**, “current,” or “instantaneous rate” associated with moving in that direction on the state space. [wikipedia](#)+1

Interpreted physically, we can choose A so that:

- $A(X)$ is the instantaneous **work** per unit time to follow the change encoded by X .
- Or the instantaneous **phase** change per unit time (like a Berry connection acting on state-space paths).
- Or an **information-theoretic cost**, e.g. rate of change of an expectation or entropy functional along X

given a small change in state $[z] \rightarrow [z + \delta z]$ represented by X , $A(X)$ quantifies how “expensive” this change is according to the chosen functional A . In FE terms, think of A as a linear form on the tangent space that you plug your vector field X into.

Section 2 – Weighted Hodge operator on 1-forms

Pick a **Hermitian** K with two features:

1. Non-degenerate spectrum (to avoid trivial symmetries).
2. A block structure that you can interpret physically—e.g. a pair of coupled qubits, or a 4-level atom with two near-degenerate subspaces.

For instance:

$$K = \begin{pmatrix} \epsilon & 0 & 0 & 0 \\ 0 & -\epsilon & g & 0 \\ 0 & g & 0 & 0 \\ 0 & 0 & 0 & \Delta \end{pmatrix}$$

with ϵ, Δ encoding energy splittings and g a coupling between levels 2 and 3.

Then define:

$$f([z]) = \langle z | K | z \rangle, \quad |z\rangle \in \mathbb{C}^4, \quad \|z\| = 1, \quad [z] \in \mathcal{M},$$

and the 1-form

$$A = df$$

on $\mathbb{CP}^3$. As we already noted, $A(X)$ for $X \in T_{[z]}\mathcal{M}$ is the instantaneous rate of change of the expectation of K along that direction of state

Now define the **geometric Hamiltonian** as an operator on scalar functions or

1-forms on $\mathbb{CP}^3$:

- Scalar version:

$$H_{\text{scalar}} = -\Delta_{\text{FS}} + \alpha f([z]),$$

with α a coupling parameter.

- 1-form version:

$$H_{\text{forms}} = \Delta_{\text{Hodge}} + \alpha f([z]) \text{Id},$$

Define the **thermal weight** at inverse temperature $\beta > 0$:

$$d\mu_\beta = e^{-\beta E([z])} d\text{vol}_{\text{FS}}.$$

We are defining this measure because we need a **geometric, physically meaningful L^2 inner product** on fields (functions or forms) that matches both the manifold's metric and the physics we're modeling (mass, potential, weights).

- The volume form $d\text{vol}_g$ turns the pointwise inner product coming from the Riemannian metric into an L^2 scalar product by integration.
- A weight (like an OU-type factor or geodesic trap) modifies the measure so that the operator you care about (Hodge–de Rham, OU/Witten Laplacian, Hamiltonian with potential) becomes self-adjoint on the corresponding weighted L^2 space.

At each p , g_p gives an inner product on $T_p M$ and hence on $T_p^* M$

$T_p^* M$ is the dual of $T_p M$: its elements are linear maps $T_p M \rightarrow \mathbb{R}$. Since g_p is non-degenerate, there is a canonical isomorphism (the flat/sharp map):

$$b: T_p M \rightarrow T_p^* M, \quad v \mapsto v^b := g_p(v, \cdot).$$

And Its inverse is $\# = b^{-1}$

This lets you transfer the inner product from $T_p M$ to $T_p^* M$: given two covectors $\alpha, \beta \in T_p^* M$, define $\langle \alpha, \beta \rangle_{T_p^* M} := g_p(\alpha^\#, \beta^\#)$. In coordinates, if (g_{ij}) is the inverse metric matrix, then

$$\langle dx^i, dx^j \rangle_{T_p^* M} = g^{ij}(p)$$

- In coordinates with g_{ij} and inverse g^{ij} , if $\alpha = \alpha_i dx^i$, $\beta = \beta_j dx^j$, then

$$\langle \alpha, \beta \rangle_p = g^{ij}(p) \alpha_i(p) \beta_j(p).$$

This gives you a scalar at each point, exactly like the dot product of vectors but using the metric to raise indices.

L^2 inner product on 1-forms

To get the global “dot product” you usually use in your weak formulations and mass matrices, you integrate the pointwise inner product against the chosen measure (usually $d\text{vol}_g$).

- The L^2 inner product of 1-forms is

$$\langle \alpha, \beta \rangle_{L^2} = \int_M \langle \alpha, \beta \rangle_p d\text{vol}_g(p).$$

In local coordinates this is

$$\langle \alpha, \beta \rangle_{L^2} = \int_M g^{ij}(x) \alpha_i(x) \beta_j(x) \sqrt{\det g(x)} dx^1 \cdots dx^n$$

Then using our Fubini-Study metric on $\mathcal{M}$ Define the weighted L^2 inner product on 1-forms:

$$\langle A, B \rangle_{L^2(\mu_\beta)} = \int_{\mathcal{M}} \langle A, B \rangle_{FS} e^{-\beta E([z])} d\text{vol}_{FS}.$$

On $\mathbb{CP}^3$ with the Fubini-Study metric g_{FS} , the pointwise inner product $\langle A, B \rangle_{FS}([z])$ of smooth 1-forms A, B is a smooth function on $\mathbb{CP}^3$.

The energy function $E([z])$ is smooth and real-valued, and the weight $e^{-\beta E([z])}$ is positive and smooth for all $[z]$ and $\beta \in \mathbb{R}$. The Fubini-Study volume form $d\text{vol}_{FS}$ is a smooth positive measure on the compact manifold $\mathbb{CP}^3$.

Putting these together, the integrand

$$\langle A, B \rangle_{FS}([z]) e^{-\beta E([z])}$$

is a smooth function, and the integral

$$\langle A, B \rangle_{L^2(\mu_\beta)} = \int_{\mathbb{CP}^3} \langle A, B \rangle_{FS}([z]) e^{-\beta E([z])} d\text{vol}_{FS}([z])$$

is finite for all smooth 1-forms B because $\mathbb{CP}^3$ is compact and the weight $e^{-\beta E}$ is bounded above and below by positive constants.

Bilinearity, symmetry, and positivity

For smooth 1-forms A, B, C and scalars λ, μ , linearity of the integral and of $\langle \cdot, \cdot \rangle_{FS}$ gives:

- $\langle \lambda A + \mu B, C \rangle_{L^2(\mu_\beta)} = \lambda \langle A, C \rangle_{L^2(\mu_\beta)} + \mu \langle B, C \rangle_{L^2(\mu_\beta)}$.
- $\langle A, B \rangle_{L^2(\mu_\beta)} = \langle B, A \rangle_{L^2(\mu_\beta)}$ because $\langle A, B \rangle_{FS}$ is symmetric pointwise.
- $\langle A, A \rangle_{L^2(\mu_\beta)} = \int_{\mathbb{CP}^3} \langle A, A \rangle_{FS} e^{-\beta E} d\text{vol}_{FS} \geq 0$, and it is zero if and only if $A = 0$ as a 1-form, since $\langle A, A \rangle_{FS}([z]) > 0$ wherever $A([z]) \neq 0$ and the weight and volume form are positive.

Thus $\langle \cdot, \cdot \rangle_{L^2(\mu_\beta)}$ is an inner product on the space of smooth 1-forms $\Omega^1(\mathbb{CP}^3)$.

Hilbert space via completion

The inner product defines a norm

$$\|A\|_{L^2(\mu_\beta)}^2 = \int_{\mathbb{CP}^3} \langle A, A \rangle_{FS} e^{-\beta E} d\text{vol}_{FS},$$

which is equivalent to the standard L^2 norm defined with respect to $d\text{vol}_{FS}$ alone, because the weight $e^{-\beta E}$ is bound above and away from zero on the compact manifold. Therefore, the metric topology induced by $\|\cdot\|_{L^2(\mu_\beta)}$ is the same as the usual L^2 topology on 1-forms, and the completion of $\Omega^1(\mathbb{CP}^3)$ under this norm is a Hilbert space $L^2(\mathbb{CP}^3, \Lambda^1 T^* \mathbb{CP}^3; \mu_\beta)$.

Recall the Hodge–de Rham Laplacian on 1-forms for the unweighted metric:

$$\Delta_1 A = d\delta A + \delta dA,$$

where δ is the codifferential defined via the Hodge star of g_{FS} .

Changing from $d\text{vol}_{FS}$ to $d\mu_\beta = e^{-\beta E} d\text{vol}_{FS}$ modifies the adjoint of d by adding a **drift term** proportional to ∇E .

Let $\mu_\beta = e^{-\beta E} \text{dvol}_{FS}$ on $\mathbb{CP}^3$, for smooth forms α and ω , the adjoint δ_β of d with respect to the weighted inner product $\langle \cdot, \cdot \rangle_{L^2(\mu_\beta)}$ satisfies

$$\int_{\mathcal{M}} \langle d\omega, \alpha \rangle_{FS} e^{-\beta E} \text{dvol}_{FS} = \int_{\mathcal{M}} \langle \omega, \delta_\beta \alpha \rangle_{FS} e^{-\beta E} \text{dvol}_{FS}.$$

To relate δ_β to the usual codifferential δ (adjoint with respect to dvol_{FS}), write

$$\int_{\mathcal{M}} \langle d\omega, \alpha \rangle_{FS} e^{-\beta E} \text{dvol}_{FS} = \int_{\mathcal{M}} \langle d\omega, e^{-\beta E} \alpha \rangle_{FS} \text{dvol}_{FS}.$$

By definition of δ (unweighted adjoint of d) we have

$$\int \langle d\omega, e^{-\beta E} \alpha \rangle_{FS} \text{dvol}_{FS} = \int \langle \omega, \delta(e^{-\beta E} \alpha) \rangle_{FS} \text{dvol}_{FS}.$$

Using the Leibniz rule and the fact that δ is first order and metric-dependent, the unweighted codifferential applied to $e^{-\beta E} \alpha$ splits into

$$\delta(e^{-\beta E} \alpha) = e^{-\beta E} \delta \alpha + (\text{drift term involving } \nabla E).$$

More precisely, in the standard Witten/OU setting one gets a contraction with the gradient of E .

Matching the integrals then gives the formal expression

$$\delta_\beta \alpha = \delta \alpha + \beta \iota_{\nabla E} \alpha,$$

where ∇E is the gradient of E with respect to g_{FS} and $\iota_{\nabla E}$ denotes interior product with ∇E .

This $\beta \iota_{\nabla E}$ is the **drift operator** associated with the weight $e^{-\beta E}$: it is the first-order term that appears in the weighted Hodge Laplacian

$$\Delta_\beta = d \delta_\beta + \delta_\beta d = \Delta + \beta (d \iota_{\nabla E} + \iota_{\nabla E} d),$$

so relative to the unweighted Laplacian Δ you have an extra drift generated by the vector field ∇E .

Define a **weighted Hodge operator** on 1-forms:

$$\mathcal{H}_\beta A := \Delta_\beta A + V([z]) A = \Delta_1 A + \beta \mathcal{L}_{\nabla E} A + V([z]) A$$

Where, $(\mathcal{L}_{\nabla E} A)_j = (\nabla E)^i \nabla_i A_j + A_i \nabla_j (\nabla E)^i$

SECTION 3: SOLVING THE EIGENVALUE PROBLEM:

While the base manifold $\mathbb{CP}^3$ is finite-dimensional (complex dimension 3, real dimension 6), the state space of forms $L^2(\mathbb{CP}^3, \Lambda^1 T^* \mathbb{CP}^3; \mu_\beta)$ is an infinite-dimensional Hilbert space. Therefore,

analyzing the spectrum of H_β is a problem in advanced operator theory and functional analysis—not standard multivariable calculus. Because H_β is a second order, heavily elliptic differential operator on a compact manifold, its spectrum will be discrete, bounded below, and the eigenforms will be smooth, provided $E([z])$ and $V([z])$ are smooth

Before we devise a strategy of how to about solve this, let's talk about the significance of solving the eigenvalue problem:

Mathematically:

The solutions to this eigenvalue problem provide a spectral decomposition of the weighted cohomology of $\mathbb{CP}^3$. The zero modes (eigenforms with $\lambda = 0$) correspond to the Betti numbers of the manifold, deformed by Morse inequalities governed by $E([z])$. The eigenforms describe the basis of geometric currents that are invariant under the thermal diffusion generated by the modified Laplacian.

Physically:

- **The Spectral Gap (λ_1):** The lowest non-zero eigenvalue dictates the *thermalization rate* (the relaxation time). It tells you exactly how fast an arbitrary initial distribution of pure states will decay into the thermal Gibbs distribution over the two-qubit space.
- **The Eigenforms:** The low-lying eigenforms represent the "slowest" decaying modes of the open quantum system. If you evaluate the 1-form A along a tangent vector X representing an increase in entanglement, the decay of this mode tells you how resilient that entanglement is against thermal noise.
- **Quantum Information Cost:** Because a 1-form assigns a linear cost to moving in a direction X , the eigenforms represent the fundamental, independent channels of "information flow" or entropy production as the two-qubit system evolves.

Can I use these types of equations to analyze or "solve" a quantum circuit?

Yes, there is a profound link between this continuous geometric formulation and discrete quantum circuits, particularly in the realm of **Variational Quantum Algorithms (VQAs)** and **Quantum Optimal Control**.

A quantum circuit is a discrete sequence of unitary gates, which traces out a piecewise-continuous path on $SU(4)$, projecting down to a path on $\mathbb{CP}^3$.

- **Gradient Descent and VQE:** In a Variational Quantum Eigensolver (VQE), a circuit is parameterized to find the ground state of a Hamiltonian K . The continuous limit of this

parameterized circuit executing gradient descent is exactly the flow generated by the vector field $-\nabla E([z])$.

- **Simulating the Heat Equation:** You can map the continuous thermal diffusion (governed by your modified Laplacian) into a quantum circuit by simulating open system dynamics. Techniques like the Quantum Imaginary Time Evolution (QITE) algorithm effectively compile continuous, non-unitary thermalization flows into a recursive sequence of unitary gates and measurements.
- **Linkage:** By solving your eigenvalue problem, you identify the exact geometric paths of least resistance (the lowest eigenforms) toward the thermal state. You can theoretically use these specific differential forms to inform the optimal unitary ansatz (the circuit architecture) needed to simulate dissipation or target specific entangled states, drastically reducing the circuit depth required.

How should we go about solving the eigenvalue problem?

Solving the eigenvalue problem

$$H_\beta A = \lambda A$$

on $\mathbb{CP}^3$ requires a blend of harmonic analysis and perturbation theory. Here is the rigorous roadmap:

- **Step 1: Exploit Symmetries (Harmonic Analysis).** $\mathbb{CP}^3$ is a homogeneous space isomorphic to $SU(4)/U(3)$. If your Hamiltonian K (and thus the energy function $E([z])$ and potential $V([z])$) possesses symmetries. For instance, if it is invariant under local unitary operations $SU(2) \otimes SU(2)$ —you can decompose the Hilbert space of 1-forms into irreducible representations of these Lie groups.
- **Step 2: The Unweighted Basis.** The spectrum of the standard Hodge Laplacian Δ_1 on $\mathbb{CP}^n$ is known. You can construct a basis of eigenforms using representation theory (often expressed via Jacobi polynomials or Wigner D-matrices lifted to the manifold).
- **Step 3: Perturbation Theory (Small β).** If the system is in a high-temperature regime ($\beta \ll 1$), the drift term $\beta \iota_{\nabla E} \alpha$ acts as a perturbation. You can apply Rayleigh-Schrödinger perturbation theory to the known spectrum of Δ_1 to find the shifted eigenvalues and eigenforms.
- **Step 4: Witten Deformation / Morse Theory (Large β).** In the low-temperature (semiclassical) limit $\beta \rightarrow \infty$, the spectrum of the Witten Laplacian localizes entirely

around the critical points of the energy landscape $E([z])$. The low-lying eigenforms concentrate around the local minima of E , and the exponentially small eigenvalues (the spectral gap) correspond to tunneling amplitudes between these vacua.